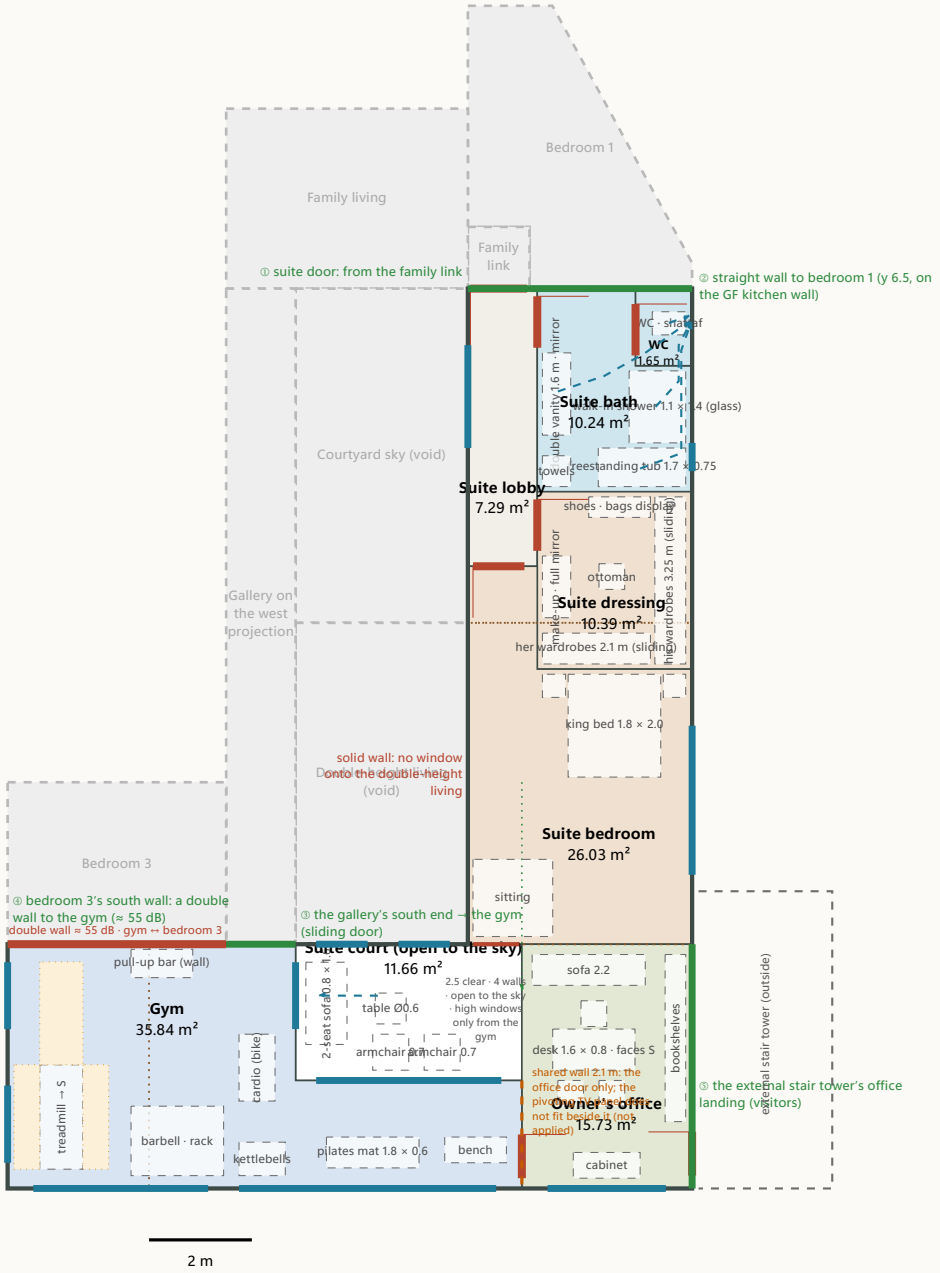
# First floor · the south wing (gym west + south, the suite court, the office east) and the master suite



first floor V6.3 · CSV coordinates (x east, y south) · wall centre lines · north up

**First floor V7.1 · part 2 of 2 (part 1: studies/ff_family.typ) · 1 Nov 2026: the gym an L on the west and the south (the treadmill on the west part), the suite court 2.5 m clear in the middle, the owner's office on the east beside the suite; the master suite unchanged**




## Rooms

| Room | Axes w × d (m) | Axes m² | ≈ clear m² |
|---|---|---|---|
| Suite lobby | 1.35 × 5.4 | 7.29 | 5.94 |
| Suite bath | polygon | 10.24 | 8.79 |
| Suite WC (closet) | 1.1 × 1.5 | 1.65 | 1.09 |
| Suite dressing | 3.01 × 3.45 | 10.39 | 9.01 |
| Suite bedroom | polygon | 26.03 | 23.56 |
| Suite court (open to the sky) | 4.4 × 2.65 | 11.66 | – |
| Owner's office | 3.31 × 4.75 | 15.73 | 13.91 |
| Gym | polygon | 35.84 | 32.52 |
| **Zone** | | **118.83** | |

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, nothing on open edges), corners ignored.

**✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the lobby route and the gym walkways · the treadmill on the west part with its manufacturer zone (0.5 m each side, 2 m free behind), the gym door free · the suite court 2.5 m clear with its furniture test · doors reached**

## Key numbers

- **Suite:** lobby 7.29 · bath 10.24 + WC closet 1.65 · dressing 10.39 · bedroom 26.03 = **55.6 m² indoors** + the suite court 11.66 m².
- **Gym** 35.84 m² on axes (≈ 32.52 clear) · **suite court** 11.66 m² (4.2 × 2.5 clear) · **owner's office** 15.73 m² (≈ 13.91 clear).
- **Walked:** suite door → bedroom door ≈ 5.4 m · bedroom door → bath door ≈ 5.5 m · bed → court ≈ 3.9 m.
- **Interfaces with the family part (green, fixed):** ① suite door at the end of the family link; ② the straight wall to bedroom 1 on y 6.5; ③ the gallery's south end: the gym's sliding door; ④ bedroom 3's south wall: the double wall to the gym; ⑤ the external stair tower's landing: the visitors' door to the office.

Plan: red = doors (thin = sliding / no leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below · grey = the family part (fixed) · dashed boxes = furniture and fixtures.

# Master suite, suite court, gym and office · rooms, drains, walls, questions

## Suite lobby · 1.35 × 5.4 m

- The suite door (from the family link) opens into a distributor: bath, dressing and bedroom each open off it; nobody walks through the dressing. A window onto the courtyard on its west wall.
- **Bent entry:** the bedroom door is at the lobby's south end, round the corner of the dressing: the bed is not seen from the suite door.

## Suite bath (luxury, compact) · 11.89 m² incl. its WC closet

- **Freestanding tub** along the south wall by an east window; **walk-in glass shower** on the east wall; **double vanity** on the west (lobby) wall facing the window light; **towel cupboard**; open floor in the middle.
- **Door** from the lobby at the north end of the west wall, opening in, swing clear; the north wall stays clear as the path from the door to the WC closet.
- **WC in its own closet** (1.1 × 1.5 m, sliding door) in the north-east corner, at the drain stack, with a shattaf.

## Suite dressing (walk-in) · 3.01 × 3.45 m

- Door from the lobby at the north end of its west wall. **His wardrobes** along the east wall and **her wardrobes** along the south wall, both sliding; a shoe and bag display on the north wall; a make-up table with a full-length mirror on the west wall; an ottoman. No window: good lighting inside the wardrobes.

## Suite bedroom · 26.03 m²

- **King bed** 1.8 × 2.0 on the dressing wall, facing south to the suite court; bedside tables; a sitting corner on the solid west wall.
- **West wall solid:** the double-height living is behind it, no window (checked). Windows east and the sliding door to the court (south).

## Suite court · 4.4 × 2.65 m on axes, 4.2 × 2.5 clear, walled on four sides, open to the sky

- 11.66 m² on axes (10.5 m² clear), **2.5 m deep clear** (owner's priority 1), from the bedroom only (a 0.9 m sliding door at x 9.05–9.95, the only door). Walled on all four sides: north the double-height living (the clerestory windows) and the suite bedroom, west the gym's west part (x 5.6, one high window, sill +1.8), south the gym's south arm (y 21.9, a 100 mm partition with high windows, sill +1.8), east the office (x 10.0, solid, no window).
- **Furniture test:** a 2-seat sofa (0.8 × 1.6) along the west wall, two armchairs (0.7) against the south wall, a table Ø0.6 between them: seating for four, with a free strip ≥ 0.75 m from the door south along the east wall and west along the north wall (checked, no overlap).
- **Clerestory windows** of the double-height living (in its south wall y 19.25): their sill stays above eye level from the court floor (or obscured glass). The gym's south arm roof edge (R6, y 21.9) is a **railing, not a parapet**, to keep the winter sun on them: with the court 2.65 m on axes the winter noon sun now reaches the whole window height (light check on design_ff.pdf).
- **Open to the sky:** waterproofing on the men's majlis roof slab, the floor falling **west** to its drain; the clerestory windows and the bedroom door open for air in summer.

## Gym · an L, 35.84 m² on axes (west part 5.6 × 4.75 + south arm 4.4 × 2.1), ≈ 32.52 m² clear

- **Owner's decision (1 Nov 2026):** the gym takes the west part (x 0–5.6) and the south arm along the facade (y 21.9–24, 1.93 m clear deep: narrowed from 2.75 to give the court its 2.5 m); the sliding door from the gallery end at its north-east corner; windows west (two) and the south band x 0.5–9.5; onto the court **high-level windows only** (sill +1.8).
- **Treadmill** (Life Fitness T5 class (assumed): 2.03 × 0.84 m) on the west part, **facing south**, 0.5 m clear each side and **2 m free behind it** (the manufacturer-type zone), 0.5 m off the west outer wall face: the slab's west edge over the foyer's load-bearing west wall, the stiffest strip of the wing, with a thickened rib B19 (x 1.05) under it; a floating floor (or dense rubber tiles) and anti-vibration pads (X6); no rigid contact with the walls.
- **Acoustics (mandatory):** the **double wall to bedroom 3** (y 19.25, x 0–4.25, red): 200 mm dense block + an independent stud lining with mineral wool, two layers of acoustic board, target **Rw ≈ 55 dB** airborne (≈ 50 dB on site), sealed at the top, no sockets back to back; the gallery's sliding door is the gym's only opening to the family part. The gym's south arm shares 2.1 m of wall with the office (200 mm block, ≈ 50 dB): the **pivoting TV panel is NOT applied** there: 0.3 m from R6's bearing + a 1.45 m panel + the 0.85 m door need ≈ 2.9 m of wall; the options are in the report (a fixed TV on each side, or a court 2.0 m deep as in the study).
- **Also:** pull-up bar on the north wall, barbell + rack and kettlebells at the south-west, the cardio bike by the gallery door, the pilates mat and a bench in the south arm; a 1.0 m walkway from the gallery door past the court's walls to the office door (checked, nothing in it).

## Owner's office · 3.31 × 4.75 m on axes, 15.73 m² (≈ 13.91 m² clear) · east, beside the suite

- Over the men's majlis roof (east end); its east wall is the outer wall, its west wall x 10 stands on the rib B21. The desk faces the south window (x 10.5–12.8); a sofa on the suite wall (quiet: the office, not the gym, is the suite's neighbour now); bookshelves on the east wall; a cabinet south-west.
- **Doors:** from the gym's south arm (0.85, hinged, opening into the office: free exit from the gym side with a **thumb-turn**, **key only** from the office side, so visitors cannot enter the house) and the **visitors' door** in the east wall (y 22.9–23.75) from the external stair tower's landing at +3.60 (a thumb-turn inside). No door onto the court. No east window (the tower covers the east wall).

## Drains (blue dashed) · no east duct

- **Suite bath:** all fixtures drain east in a **sunken bath floor** to one **stack on the outside face of the east wall**, down to the east-yard manhole where the kitchen already drains. Nothing hangs in the GF kitchen ceiling; the hood on the GF east wall stays clear of the stack (south of y ≈ 7.5).
- No drain over the master bedroom. The suite court drains to its own floor drain.

## Walls (engineer)

- **On GF lines:** y 6.5 (bedroom 1 / suite: the GF kitchen's north wall), the outer walls, x 8.95 (courtyard / double-height living). The suite bedroom spans the GF kitchen / living wall (y 13.0) with no wall above it.
- **Off the GF lines:** x 10.3, y 10.45, x 12.211 and y 8.0, y 11.9 and y 13.9, all over the GF kitchen; on the men's majlis roof (1 Nov 2026): x 5.6 (gym west part / court) on the rib B20, **x 10** (court / office and gym / office) on a new full-length rib B21, y 21.9 (court / gym south arm) a 100 mm partition across the ribs; the court's roof edge R6 (4.4 m) on the two walls: **X5 revised, not verified** (studies/structure.typ).
- **Loads and vibration** of the gym (treadmill) and the office over the men's majlis: engineer.

## Questions for the owner

1. **Bedroom:** king bed facing the court, or another arrangement (TV wall, sitting area)?
2. **Suite court:** 11.66 m², 2.5 m clear with seating for four: confirm the furniture; clerestory privacy (sill height or obscured glass) to settle.